Dylan Thornburg

1.

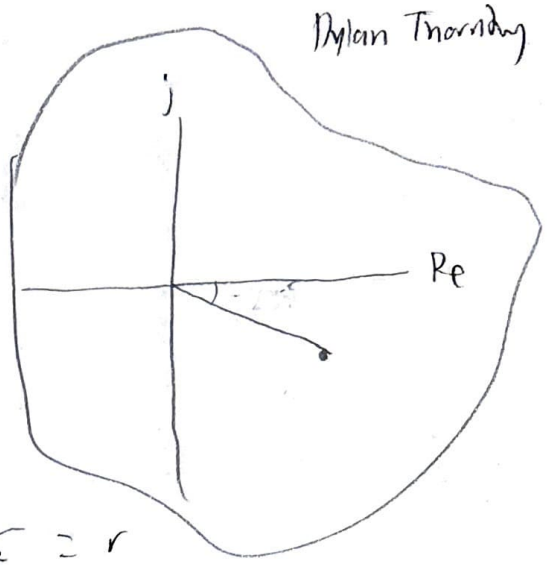
$$\frac{6\angle 45^\circ}{2.5+11j} + \frac{\angle 15^\circ}{10}$$

$$6\cos 45 + j6\sin 45 + \cos 15 + j\sin 15$$

$$\frac{4.243 + j4.243}{2.5+11j} + \frac{.966 + j.259}{10}$$

$$\frac{42.43 + j42.43 + 2.5+11j(.966 + j.259)}{25+110j}$$

$$.547 - .258j$$



$$\sqrt{.547^2 + (-.258)^2} = r$$

$$r = .605$$

$$\tan^{-1}\left(\frac{-.258}{.547}\right) = -25.252$$

rect: $.547 - .258j$
polar: $.605\angle -25.252^\circ$
exp: $.605e^{-25.252j}$

2.

$$\frac{2+j3}{1-j6} + \frac{7-j8}{-5+j11}$$

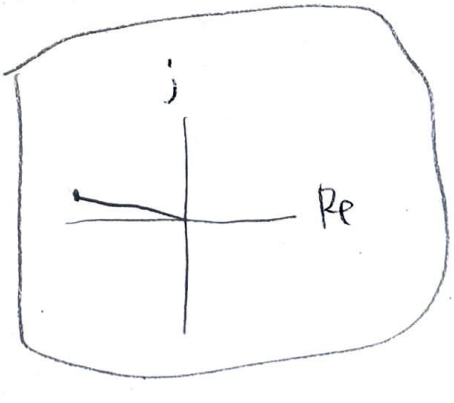
$$\frac{3.606\angle 56.310^\circ}{6.083\angle -80.538^\circ} + \frac{10.630\angle -48.814^\circ}{12.083\angle 114.444}$$

$$.592\angle 136.848 + .880\angle -163.258$$

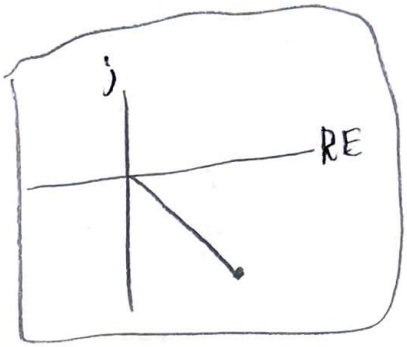
$$-.432 + j.405 + -.843 - j.253$$

$$-1.275 + .152j$$

rect: $-1.275 + .152j$
polar: $1.284\angle 173.202^\circ$
exp: $1.284e^{173.202j}$



3. $$\frac{\left(\frac{10+j20}{5\angle 53.1}\right)^2 (16-20j)}{10+j5}$$



$5\angle 53.1$

$3 + 3.998j$

rect: $24.796 - 38.528j$

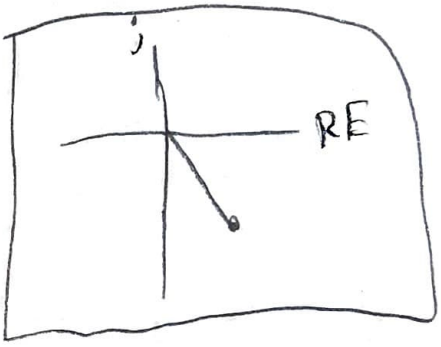
polar: $45.818\angle -57.235^\circ$

exp: $45.818e^{-57.235j}$

4.

$32\angle(-20)$

$30.070 + -10.945j$



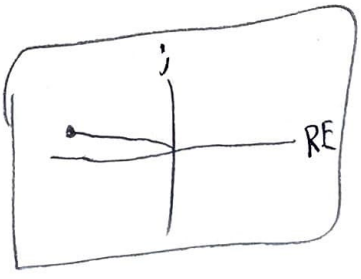
rect: $.415 - .628j$

polar: $.753\angle -56.542^\circ$

exp: $.753e^{-56.542j}$

5.

rect: $-.766 + .139j$

polar: $.779\angle 169.715^\circ$

exp: $.779e^{169.715j}$